Supplementary Information for: Hard thermodynamic constraints and the temperature extrapolation of learned water isotherms in metal-organic and covalent organic frameworks: a pre-registered test

Abdulhamid Batayhi

Supplementary Table S1: Data

Material	Class	Set	Data	Source DOI	T range (K)	n_T	Points per T	p/p_{sat} range	q_{ref} (mol kg^{-1})
COF-432	COF	conf.	vector	10.1021/jacs.9b13094	283.15–313.15	3	28–29	0.063–0.870	16.04
COF-482-hydrazide	COF	conf.	raster	10.1021/acscentsci.2c00398	288.15–308.15	3	25–51	0.08–0.898	34.70
COF-SO3H	COF	conf.	vector	10.1002/cssc.202301906	298.15–318.15	3	18–18	0.0003–0.900	17.04
HFPTP-DMePDA-COF	COF	conf.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	9–18	0.39–0.897	7.91
HFPTP-PDA-COF	COF	conf.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	24–32	0.0041–0.898	16.78
NO-PI-3-COF	COF	conf.	vector	10.1021/jacs.3c02572	288.15–308.15	3	30–37	0.0091–0.882	12.74
PI-3-COF	COF	conf.	vector	10.1021/jacs.3c02572	288.15–308.15	3	28–37	0.01–0.887	24.80
TFBCz-PDA-COF	COF	conf.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	29–30	0.0023–0.899	27.26
Al-fumarate	MOF	conf.	raster	10.1021/acscentsci.9b00745	283.15–323.15	3	56–59	0.049–0.892	23.86
CALF-25	MOF	conf.	ISODB	10.1021/ja306812r	298.00–313.00	3	17–20	0.018–0.894	3.75
Co2Cl2(BTDD)	MOF	conf.	vector	10.1021/acscentsci.7b00186	283.00–298.00	3	31–34	0.0072–0.862	53.97
Mg-SU-102	MOF	conf.	numeric	10.1038/s41467-025-59551-2	288.15–328.15	9	21–21	0.0013–0.819	21.11
Zn2(bptc)	MOF	conf.	ISODB	10.1021/ja060946u	288.00–318.00	4	25–33	0.023–0.885	8.38
TFPPy-PDA-COF	COF	dev.	raster	10.1038/s41467-021-27128-4	283.15–298.15	3	7–15	0.46–0.900	35.41
CuBTC	MOF	dev.	ISODB	10.1016/j.ijheatmasstransfer.2012.07.068	293.00–358.00	9	23–41	0.0029–0.508	25.29
MIL-100	MOF	dev.	ISODB	10.1016/j.ijheatmasstransfer.2012.07.068	293.00–358.00	9	25–35	0.011–0.603	18.11
MIL-160	MOF	dev.	ISODB	10.1002/adma.201502418	293.00–313.00	3	32–54	0.0052–0.884	21.79

Supplementary Table S1. The 13 confirmatory materials (conf.) and the four development materials (dev.) after the inclusion rules and the preprocessing. Data: how the points were obtained (vector or raster digitisation of the published figure, numeric source data, or the NIST adsorption database, ISODB). n_T: number of temperatures; points per temperature with $p/p_{\mathrm{sat}} \leq 0.90$; q_{ref}: the largest uptake over all temperatures in this range, used only for scoring. Every decision and its reason is in data/inclusion_log.csv.

Supplementary Note 1: Can each verdict fail? Synthetic worlds and power

Before the freeze, the whole pipeline (the tuned configurations, fitting from 5 seeds, the violation counts and the analysis code) was run in synthetic worlds that share no uptake data with the confirmatory set. Each world has 13 materials; material i uses the temperatures of confirmatory material i. An expected outcome was written for each world before it ran. In KS0 the truth is a two-site Sips isotherm with van't Hoff affinities, inside the family of P; in DA it is a Dubinin-Astakhov curve, inside the family of the hard characteristic curve (A-hard); in FW it is a Kirchhoff-Sips isotherm with a large heat capacity, so that van't Hoff is wrong; FLAT has one weakly curved Langmuir site; SUB has a sub-linear Sips isotherm outside the family of P and inside that of the penalty model V-pen.

The comparison of P with A-hard discriminated in both directions: P extrapolated better in KS0 ($R = 0.52$, exact 95 % interval 0.35 to 0.70) and A-hard better in DA ($R = 1.26$, 1.07 to 1.49). Three expectations were missed. In FW the comparison of P with the free temperature network F-hard still read "P better" ($R = 0.33$): with two training temperatures the free network cannot extrapolate, so this comparison cannot fail and is reported descriptively. In FLAT the unconstrained network F-none violated physics "often". In SUB, built for V-pen to win, P was slightly better ($R = 0.96$, 0.94 to 0.99); the mixture of Langmuir terms in P approximates a sub-linear isotherm over the measured range. The accuracy comparison of hard constraints with penalties therefore could not be made to fail synthetically, and it is tested as non-inferiority.

The power of each verdict was simulated with the frozen analysis code at $n = 13$ (2000 simulations per setting). At the spread expected for the comparison with V-pen (0.158 decades), the non-inferiority criterion is met with probability 0.0225 when the true ratio is at the margin, $R = 1.25$, and 0.51 (normal) or 0.74 (Student t, 3 degrees of freedom) when the two models are equally accurate. A "not met" outcome is therefore likely even under equal accuracy and is never read as evidence that the constraints cost accuracy.